

Fold-change detection as a chemotaxis model discrimination tool

Abdullah Hamadeh*, Brian Ingalls*, Eduardo Sontag^{†,‡}

Abstract—Fold-change detection (FCD) is the property that a dynamical system with an adapting output will exhibit identical transient output responses when its input signals are scaled. This feature was recently demonstrated in the chemotactic response of the bacterium *Escherichia coli*, confirming earlier theoretical predictions. The chemotaxis pathway of the bacterium *Rhodobacter sphaeroides* has the same modular structure as that in *E. coli* but is significantly more complex in that it has multiple homologues of the latter’s chemotaxis proteins and features two, rather than one, chemosensory cluster. Recent experimental results suggest that *R. sphaeroides* may also exhibit FCD.

In this paper, we present a set of theoretical assumptions on the dynamics of the *R. sphaeroides* chemosensory system, and use these to fit an integrated chemotaxis model to experimental data. We then show that the assumptions we place are sufficient to make FCD a robust property of this chemotaxis pathway, in agreement with preliminary experimental evidence. We argue that the fact that the model we present here is able to reproduce this transient dynamic property whilst earlier models cannot makes FCD a useful tool for model discrimination on the basis of transient dynamic response. This is in contrast to earlier model discrimination methods which tested the validity of models based on their ability to reproduce a finite set of experimental data. Further experiments that can provide additional validation of our theoretical assumptions are suggested.

I. INTRODUCTION

Mathematical modeling has played an important role in explaining key features of biological systems, such as their response to stimuli and their robustness properties. These advances have traditionally been realized through a cyclical process of model construction and validation, the latter typically involving a comparison of simulated outputs with experimental observation.

This paper gives an overview of a complementary general approach to model falsification, based on the use of the qualitative properties of the transient responses of dynamical systems with adapting outputs, as described in [1]. We first present a model of the chemotaxis pathway of the bacterium *Rhodobacter sphaeroides*. Departing from previous representations of this system, this model assumes that the bacterium’s receptor dynamics can be approximated using the Monod-Wyman-Changeaux (or, MWC) allosteric model [2], as has recently been done with *Escherichia coli* chemotaxis models [3], [4], [5]. It is shown that this model can reproduce previously published experimental time-series data.

We then discuss a general mathematical result given in [1] that shows that in certain concentration ranges, scaling the ligand input signals detected by the bacterium leads to the same chemotactic output response, a behavior termed *fold change detection* (FCD), recently observed experimentally in *E. coli*. Recent experimental results in [6] suggest that *R. sphaeroides*, like *E. coli*, also exhibits this behavior. Unlike previous *R. sphaeroides* chemotaxis models, we show, both analytically and through simulations, that the model we present here is capable of displaying FCD. In this way, we are able to invalidate previous models through testing for a characteristic transient response that does not rely on fitting to a fixed set of time-series data.

We additionally discuss how the assumptions we place on this model make FCD a feature of the chemotactic response of this bacterium that is robust to substantial variation in the pathway’s signaling structure. This is in line with experimental results also presented in [6] that FCD appears to feature in bacteria that have different cell architectures arising from different growth conditions.

II. BACKGROUND

The chemotaxis pathways of the bacteria *E. coli* and *R. sphaeroides* share homologies and a fundamental modular structure illustrated in Figure 1. In both, receptor proteins are responsible for sensing the presence of ligands in their local environment. These signals are then transduced through phosphorylation to a cascade of chemotaxis proteins, and the net effect of ligand sensing is a reduction in this kinase activity of the receptors. The protein CheY in *E. coli* and its homologues CheY₃, CheY₄ and CheY₆ in *R. sphaeroides*, when phosphorylated, interact with the bacteria’s flagellar motors in a way that increases their stopping rates (see [7]). At the same time, protein CheB in *E. coli* and its homologues CheB₁ and CheB₂ in *R. sphaeroides* act as methylsterases. De-methylation of the receptors by the CheB proteins has the effect of reducing the receptors’ kinase activity, and thus the CheB proteins in the two bacteria can be viewed as negative feedback signals. Viewed holistically, this feedback system causes the bacteria to climb attractant ligand gradients: a sensed increase in attractant concentration causes the amount of phosphorylated CheY and CheB protein to fall. The fall in CheY proteins leads to a lengthening of the bacterium’s average swim length (due to a decrease in its stopping frequency). However, the fall in CheB proteins reduces the de-methylation rate at the receptors, thus causing a resurgence in their kinase activity and a restoration of the pre-stimulus steady state stopping rate of the flagellar motors [8]. This return to the steady state stopping frequency

* Department of Applied Mathematics, University of Waterloo, Waterloo, Ontario, Canada. Emails: ahamadeh@uwaterloo.ca (Abdullah Hamadeh), bingalls@uwaterloo.ca (Brian Ingalls)

[†]Department of Mathematics, Rutgers University, Piscataway, New Jersey, USA. Email: sontag@math.rutgers.edu

[‡]Corresponding author.

is termed adaptation, and has been studied extensively and modeled as a phenomenon that arises from a biochemical mechanism that parallels integral control [9], [10], [7].

The chemotaxis pathway in *R. sphaeroides* differs from that in *E. coli* in one key respect however: in addition to the set of chemosensory receptors that span the cell membrane in the two bacteria, *R. sphaeroides* features an additional set of receptors found within its cytoplasm, and it is believed that, in addition to integrating metabolic information from within the cell, information about the external ligand concentration is conveyed to this internal set of receptors [7].

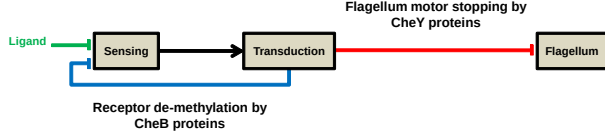


Fig. 1. Main modules of the *E. coli* and *R. sphaeroides* chemotaxis pathways.

As in [11], [12], [13], the dynamics of a single receptor cluster can be modeled as a first-order system. Since *R. sphaeroides* has two such clusters, its sensing dynamics can be modeled as two such systems that are coupled. The average degree of methylation of the membrane receptors is denoted by the state m , and its kinase activity, a (the ‘output’ of this system in control theoretic terminology) is a static function of m and the ligand concentration sensed by the membrane receptors, L (the ‘input’). Analogously, at the cytoplasmic cluster, we have methylation state $\tilde{m}$, a kinase activity output $\tilde{a}$ and cytoplasmic sensed ligand input $\tilde{L}$. For simplicity, we assume that $\tilde{L} = L$, although this restriction is relaxed in [1] (in the remainder of this paper, signals associated with the cytoplasmic cluster will be similarly denoted with a tilde). A simple model of the receptor dynamics is then given by

$$\begin{aligned} \dot{m} &= F(a, x) & a &= G(m, L) \\ \dot{\tilde{m}} &= \tilde{F}(\tilde{a}, x) & \tilde{a} &= \tilde{G}(\tilde{m}, \tilde{L}) \end{aligned} \quad (1)$$

where x is a vector of signals that generally represents the levels of phosphorylated chemotaxis proteins. It features in the methylation dynamics due to the role of the CheB₁, CheB₂ proteins in de-methylating the receptors. As in [8], [11], the condition for adaptation, that only inactive receptors undergo methylation and only active receptors undergo de-methylation, leads to the functions F and $\tilde{F}$ being monotonically decreasing functions of a and $\tilde{a}$ respectively.

The kinase activities of the two receptors are transduced to the CheB and CheY chemotaxis proteins via the proteins CheA₂ (at the membrane cluster) and CheA₃ (at the cytoplasmic cluster): the auto-phosphorylation rates of these two proteins are modeled as being proportional to $a, \tilde{a}$ respectively. The CheY and CheB proteins then compete for phosphoryl groups from CheA₂, CheA₃: CheA₂ phosphorylates CheY₃, CheY₄, CheY₆, CheB₁ and CheB₂ whilst CheA₃

phosphorylates CheB₂ and CheY₆. The reaction rates for all of these phosphorylations are given in [14], [12]. As in [12], this phosphotransfer network is represented as a nonlinear system, with a state vector that gives the concentration of phosphorylated chemotaxis proteins:

$$x = [A_{2p} \ Y_{3p} \ Y_{4p} \ A_{3p} \ Y_{6p} \ B_{1p} \ B_{2p}]^T$$

and dynamics governed by

$$\dot{x} = H(x, a, \tilde{a}) \quad (2)$$

where $H(x, a, \tilde{a})$ in [12]. The equations (1) and (2) then fully represent the sensing and signal transduction dynamics of the *R. sphaeroides* chemotaxis system.

The precise mechanism through which the CheY proteins interact with the flagellum is unknown. Experiments show that CheY₆ and either one of CheY₃ and CheY₄ are required for chemotaxis [7]. A range of possible connectivities between the CheY proteins and the flagellum were studied in [12], where it was even suggested that CheY₃ and CheY₄ could interact with the cytoplasmic cluster, affecting its kinase activity, whilst CheY₆ stops the flagellar motors. There is additional uncertainty about the connectivity of the CheB proteins with the two sensing clusters. In [13] it is suggested that the protein CheB₁ de-methylates the membrane cluster whilst CheB₂ de-methylates both clusters, and this connectivity is used in the model presented here. It is shown in [1] that under some theoretical assumptions, which we will summarize in the next section, FCD can be displayed by the bacterium despite this uncertainty in the precise structure of the biochemical reaction network.

III. MODELING CHEMOTAXIS IN *R. sphaeroides*

A. An MWC model of receptor dynamics

In several works such as [15], [16], [17], [11], the Monod-Wyman-Changeaux allosteric model [2] has been employed to approximate chemoreceptor activity. The model starts with the assumptions that receptors can be in either an active state or an inactive state, and that ligands bind more strongly with inactive receptors than with active receptors. This latter quality is quantified by the dissociation constants between ligands and inactive receptors, $K_I, \tilde{K}_I$ being lower than the dissociation constants between ligands and active receptors $K_A, \tilde{K}_A$. The quantities $a(t)$ and $\tilde{a}(t)$ given in the previous section respectively represent the probabilities of activity of the membrane and cytoplasmic clusters at any given time. An approximation of this probability for the membrane receptors is obtained by first denoting the free energy of active receptors by E_A and of inactive receptors by E_I . From [18], [16], [11] the probability of activity is approximated by the ratio of the Boltzman factor of the active state to the sum of the Boltzman factors of the active and the inactive states:

$$a(t) = \frac{\exp(-E_A)}{\exp(-E_I) + \exp(-E_A)} = \frac{1}{1 + \exp[-E_\Delta]} \quad (3)$$

where $E_\Delta = E_I - E_A$ is the free energy difference between the active and inactive states. Similarly, for the cytoplasmic

cluster we have the approximation

$$\tilde{a}(t) = \frac{\exp(-\tilde{E}_A)}{\exp(-\tilde{E}_I) + \exp(-\tilde{E}_A)} = \frac{1}{1 + \exp[-\tilde{E}_\Delta]} \quad (4)$$

with $\tilde{E}_\Delta = \tilde{E}_I - \tilde{E}_A$. As discussed in [19], we assume there are two contributions to the free energy differences E_Δ , $\tilde{E}_\Delta$, one from the degree of methylation of receptors (quantified by functions $g_m(m)$ and $\tilde{g}_m(\tilde{m})$) and the other from the sensed ligand concentration (quantified by functions $g_L(L)$ and $\tilde{g}_L(\tilde{L})$). We further assume that these two contributions can be combined additively, so that

$$E_\Delta = -N[g_m(m) + g_L(L)] \text{ and} \\ \tilde{E}_\Delta = -\tilde{N}[\tilde{g}_m(\tilde{m}) + \tilde{g}_L(\tilde{L})]$$

with $N, \tilde{N}$ representing the degree of cooperativity arising from the clustering of chemoreceptors [18].

The contributions to the free energy differences from receptor methylation, $g_m(m)$ and $\tilde{g}_m(\tilde{m})$, are taken to be affine, decreasing functions of $m, \tilde{m}$ respectively as was shown to be the case for *E. coli* in [19], with

$$g_m(m) = \alpha(m_0 - m) \text{ and } \tilde{g}_m(\tilde{m}) = \tilde{\alpha}(\tilde{m}_0 - \tilde{m})$$

where $\alpha = \tilde{\alpha} = 2$ and $m_0 = \tilde{m}_0 = 5$.

The contribution from ligand binding is proportional to the logarithm of the ligand concentrations $L, \tilde{L}$, which arises from the fact that there is a loss in ligand translational entropy when ligands bind to the receptors [16], [11]. The loss in entropy is greater in the case of ligands binding to active receptors than to inactive receptors due to the lower affinity to the former, and this is approximated by $-\ln(\frac{L}{K_A})$ and $-\ln(\frac{\tilde{L}}{K_A})$ in the case of active receptors and $-\ln(\frac{L}{K_I})$ and $-\ln(\frac{\tilde{L}}{K_I})$ for inactive receptors. As in [16], [11], the contribution of ligand binding to the free energy differences of the receptors between their active and inactive states is then given by

$$g_L(L) = \ln\left(1 + \frac{L}{K_I}\right) - \ln\left(1 + \frac{L}{K_A}\right) \text{ and} \\ \tilde{g}_L(\tilde{L}) = \ln\left(1 + \frac{\tilde{L}}{\tilde{K}_I}\right) - \ln\left(1 + \frac{\tilde{L}}{\tilde{K}_A}\right)$$

for the membrane and cytoplasmic clusters respectively. Note that since $K_I < K_A$ and $\tilde{K}_I < \tilde{K}_A$, the greater affinity of ligands for inactive receptors makes $a(t)$ and $\tilde{a}(t)$ decreasing functions of L and $\tilde{L}$ respectively. From the [11], we adopt the values $K_I = \tilde{K}_I = 18\mu M$, $K_A = \tilde{K}_A = 3mM$.

With this framework, we now make the important observation, analogous to that in [3], that in the ligand concentration range $K_I \ll L \ll K_A$ and $\tilde{K}_I \ll \tilde{L} \ll \tilde{K}_A$, the chemoreceptor activities can be approximated by

$$a = \frac{1}{1 + \left[\exp(\alpha[m_0 - m]) \frac{L}{K_I}\right]^N} \text{ and} \\ \tilde{a} = \frac{1}{1 + \left[\exp(\tilde{\alpha}[\tilde{m}_0 - \tilde{m}]) \frac{\tilde{L}}{\tilde{K}_I}\right]^{\tilde{N}}} \quad (5)$$

This observation will be used in Section IV in determining the ligand concentration range in which FCD can be observed.

B. An integrated *R. sphaeroides* chemotaxis model

In Section III-A we outlined the physical basis for the MWC model of the chemotaxis receptors which leads to the approximation (5). Here, we combine this approximation with the sensing and signal transduction dynamics of the *R. sphaeroides* chemotaxis pathway in (1) and (2).

Following the models in [13], mass action kinetics give the following general form for $F, \tilde{F}$ in (1)

$$\dot{m} = F(a, \tilde{a}, w(\mathbf{x})) = k_R(1 - a) - k_{B_1}B_{1_p}a - k_{B_2}B_{2_p}a \\ \dot{\tilde{m}} = \tilde{F}(a, \tilde{a}, \tilde{w}(\mathbf{x})) = \tilde{k}_R(1 - \tilde{a}) - \tilde{k}_{B_2}B_{2_p}\tilde{a} \quad (6)$$

where $k_R, \tilde{k}_R > 0$ are methylation and $k_{B_1}, k_{B_2}, \tilde{k}_{B_2} > 0$ de-methylation rate constants. From Section III-A, the membrane and receptor cluster activities are given by

$$a = \frac{1}{1 + \left[\exp(\alpha[m_0 - m]) \left[\frac{1 + \frac{L}{K_I}}{1 + \frac{L}{K_A}}\right]^N\right]} \text{ and} \\ \tilde{a} = \frac{1}{1 + \left[\exp(\tilde{\alpha}[\tilde{m}_0 - \tilde{m}]) \left[\frac{1 + \frac{\tilde{L}}{\tilde{K}_I}}{1 + \frac{\tilde{L}}{\tilde{K}_A}}\right]^{\tilde{N}}\right]} \quad (7)$$

and the dynamics of the phosphorylation cascade is given by

$$\dot{x} = H(x, a, \tilde{a}) \quad (8)$$

as described in Section II, with $H(x, a, \tilde{a})$ given in [12].

Together with the simplifying assumption (which is relaxed in [1]) that $\tilde{L} = L$ and with the model of flagellar rotation inhibition by the CheY proteins given in [1], we can arrive at an integrated ODE description of the *R. sphaeroides* chemotaxis network by fitting the constants $k_R, \tilde{k}_R, k_{B_1}, k_{B_2}, \tilde{k}_{B_2}$ in (6) to experimental time-series data from [13]. A schematic is illustrated in Figure 2, and a simulation of the model together with the tethered cell assay from [13] to which the model was fitted is shown in Figure 3. The fitted parameters of the model are as follows: $k_R = \tilde{k}_R = 0.0057$ $k_{B_1} = \tilde{k}_{B_2} = 2.376$ $k_{B_2} = 2.970$, $N = \tilde{N} = 1$.

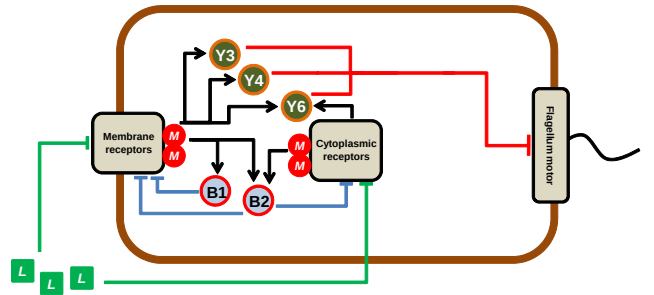


Fig. 2. Schematic of the *R. sphaeroides* chemotaxis mode

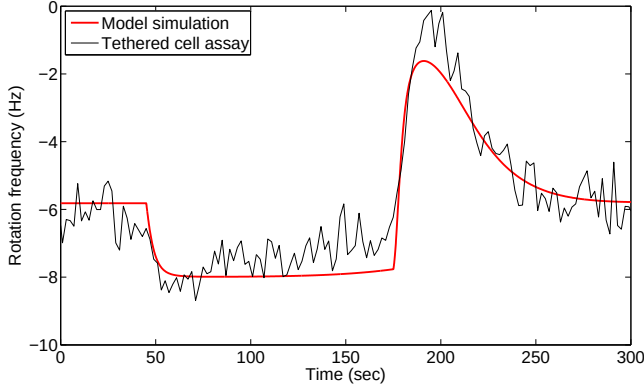


Fig. 3. Simulation of fitted model (red) in response to a step rise (at 45 seconds) and fall (at 175 seconds) in the ligand level L from $L = 0$ to $L = 100$ and back to $L = 0$, with a tethered cell assay (black).

IV. FOLD CHANGE DETECTION IN *R. sphaeroides*

The following definition of fold change detection (FCD) closely follows that given in the earlier works [3], [5].

Definition 1: The *R. sphaeroides* chemotaxis system (6), (7) and (8) exhibits FCD in response to a sensed ligand input signal $L(t)$ if its receptor activities $a(t), \tilde{a}(t)$, initially at a steady state corresponding to $L(0)$, are independent of linear scalings $p > 0$ of the input $L(t)$.

The following theorem gives conditions for FCD for the model (6), (7) and (8).

Theorem 1: Under approximation (5) the system (6), (7), (8) will exhibit FCD in the sense of Definition 1 if it has a unique steady state for any constant ligand input L .

Proof: Under approximation (5), the system (6), (7), (8), can be written as

$$\begin{aligned} \dot{m} &= F(a, x), \quad a = G(\exp(-m)L) \\ \dot{\tilde{m}} &= \tilde{F}(\tilde{a}, x), \quad \tilde{a} = \tilde{G}(\exp(-\tilde{m})\tilde{L}), \quad \tilde{L} = L \\ \dot{x} &= H(x, a, \tilde{a}) \end{aligned} \quad (9)$$

Consider a ligand input signal $L(t)$, defined on $[0, \infty)$. Suppose that (9) is initially at the unique steady state pre-adapted to the constant input $L(0)$, and exhibits the solution $\mathbf{m}_1(t) = [m(t)_1 \quad \tilde{m}_1(t) \quad x_1(t)^T]^T$ in response to the input $L_1(t) = L(t)$, yielding outputs $a_1(t)$ and $\tilde{a}_1(t)$. Now consider the solution of (9) to the input $pL(t)$ when it is initially at the unique steady state pre-adapted to the constant input $pL(0)$, $p > 0$. We denote this solution by $\mathbf{m}_2(t) = [m(t)_2 \quad \tilde{m}_2(t) \quad x_2(t)^T]^T$ and the corresponding outputs by $a_2(t)$ and $\tilde{a}_2(t)$. The solution $\mathbf{m}_2(t)$ and outputs $a_2(t)$ and $\tilde{a}_2(t)$ are equivalent to those of the system

$$\begin{aligned} \dot{m} &= F(a, x), \quad a = G(\exp(-m)pL) \\ \dot{\tilde{m}} &= \tilde{F}(\tilde{a}, x), \quad \tilde{a} = \tilde{G}(\exp(-\tilde{m})p\tilde{L}), \quad \tilde{L} = L \\ \dot{x} &= H(x, a, \tilde{a}) \end{aligned} \quad (10)$$

in response to the input $L(t)$ when its initial state is the unique steady state pre-adapted to the constant input $L(0)$. Consider now the changes of variables $\exp(-w) =$

$\exp(-m)p$ and $\exp(-\tilde{w}) = \exp(-\tilde{m})p$ so that (10) becomes

$$\begin{aligned} \dot{w} &= F(a, x), \quad a = G(\exp(-w)L) \\ \dot{\tilde{w}} &= \tilde{F}(\tilde{a}, x), \quad \tilde{a} = \tilde{G}(\exp(-\tilde{w})\tilde{L}), \quad \tilde{L} = L \\ \dot{x} &= H(x, a, \tilde{a}) \end{aligned} \quad (11)$$

Note that in response to the input $L(t)$, when its initial state is the unique steady state pre-adapted to the constant input $L(0)$, (11) will exhibit the solution $\mathbf{w}(t) = [w(t) \quad \tilde{w}(t) \quad x(t)^T]^T = \mathbf{m}_1(t)$. It will therefore exhibit outputs $a_1(t)$ and $\tilde{a}_1(t)$. However, since the outputs of (10) are $a_2(t)$, $\tilde{a}_2(t)$, it follows that $a_1(t) = a_2(t)$ and $\tilde{a}_1(t) = \tilde{a}_2(t)$. This means that the outputs arising from the scaled input will be identical to the outputs arising from the unscaled input, and hence the system shows FCD in the sense of Definition 1. ■

A more general version of Theorem 1, which applies to a much wider class of chemotaxis systems structures, is given in [1]. For example, with the assumptions on the methylation dynamics given in Section III-A, FCD can be shown for models where the connectivity between the CheB proteins and the two sensing clusters is different to that presented in this model, and where there is an interaction between the CheY proteins and the cytoplasmic cluster that alters $\tilde{a}$, as in [13]. It also shown FCD will hold when the relationship between the externally sensed ligands L and those detected by the cytoplasmic cluster, $\tilde{L}$ can be generalized to a linear, dynamic one. Such a relation could be a more realistic representation of the actual biochemical relationship between these two signals, modeling the likely propagation delays arising from internalization of the externally sensed ligand.

An interesting feature to note about (6), (7), (8) is that, under the conditions of Theorem 1, if the system exhibits FCD in its outputs $a, \tilde{a}$, then it also shows FCD in the concentrations of its phosphorylated chemotaxis proteins. This arises from the following argument: the outputs $a, \tilde{a}$ form the only inputs to (8). Regardless of the the value of any constant input $L(0)$ to which the system (6), (7), (8) is adapted, the outputs $a(0), \tilde{a}(0)$ adapt to the same steady state value. At steady state, the phosphorylation cascade (8) is adapted to the unique equilibrium corresponding to its steady state input $a(0), \tilde{a}(0)$. By Theorem 1, the inputs to (8), $a(t)$ and $\tilde{a}(t)$ do not change with a scaling of the ligand input $L(t)$. Therefore, regardless of any scaling of the ligand input $L(t)$, the phosphorylation cascade (8) has the same initial state and is excited by the same input. Its states must therefore exhibit the same output response. Figure 4 shows that in the ligand range $K_I \ll L \ll K_A$, $\tilde{K}_I \ll \tilde{L} \ll \tilde{K}_A$, where approximation (5) applies, the model (6), (7), (8) shows very approximate FCD in its phosphorylated CheY₆ concentration.

Since the flagellar activity is also a function of the CheY proteins, this means that FCD should also be observed at the flagellar rotation level. The fact that FCD can be observed in both the concentration of phosphorylated chemotaxis proteins and in the flagellar activity is useful because it suggests that a variety of different experimental methods can

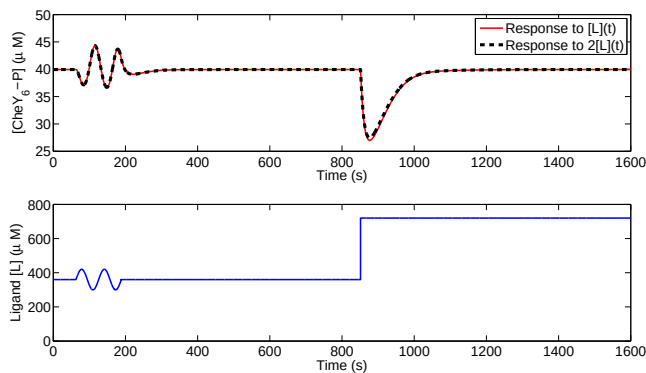


Fig. 4. Output of model in response to the ligand input $L(t)$ (red, solid) and input $2L(t)$ (black, dashed).

be used to test for this dynamic behavior, including FRET measurements and tethered cell assays. In [6], preliminary experimental evidence for FCD in *R. sphaeroides* is presented in the form of tethered cell assays that show constant adaptation times in response to five-fold step changes in ligand concentration. More conclusive evidence for FCD, such as that in [4], would show that the entire shape of the flagellar response would not change as a result of ligand input scalings.

V. DISCUSSION

Unlike previous models, such as those in [12], [13], the *R. sphaeroides* model we have presented features methylation dynamics that are based on the MWC approximation. This allosteric model of the receptor kinase activity follows that presented for *E. coli* in [11], [4]. The theorem we have presented and the more general result in [1] show that in a certain ligand concentration range, these methylation dynamics lead to FCD. In [1], it is shown that earlier models such as [13] do not show this property, even though both the model we present here and that in [13] are able to reproduce the same experimental data shown in Figure 3. There is, at present, preliminary evidence for FCD given in [6]. If more conclusive evidence such as that in [4] for FCD in *E. coli* could be demonstrated for *R. sphaeroides* then the model of the methylation dynamics we have presented in this paper and in [1] would invalidate that in [13]. What makes this approach to model discrimination different to that presented in [12], [13] is that a *transient dynamic property* is used to differentiate between the two models. In other words, the model we present here is expected to show the FCD property in response to *any* ligand input scaling that lies in the ligand range $K_I \ll L \ll K_A$, $\tilde{K}_I \ll \tilde{L} \ll \tilde{K}_A$. This is in contrast to [13] which relied on a fixed number of time-series experiments to discriminate between the models.

Another feature to note from the result presented in [1] is that FCD is a robust property of *R. sphaeroides* chemotaxis models that feature the MWC methylation dynamics presented in Section III-A. This is in line with experimental evidence in [6] that suggests that FCD may be observed in *R. sphaeroides* bacterial strains that are grown in different

conditions that lead to different stoichiometries, protein copy numbers and reaction rates in the bacterium. If the models presented here and in [1] are accurate, we would also expect FCD to appear in mutants that have gene deletions and over-expressions. Two possible experiments which could be performed to confirm this are the deletions of CheB_2 and the over-expression of CheY_4 . These mutations were shown in [12], [13] to not entirely eliminate the bacterium's chemotaxis response and would be candidates for further validation of our theoretical assumptions.

VI. ACKNOWLEDGEMENTS

Authors AH and BI would like to acknowledge the support of an Early Researcher Award from the Government of Ontario, Canada. This work was also supported in part by AFOSR grant FA9550-11-1-0247 and NIH grants 1R01GM100473 and 1R01GM086881.

REFERENCES

- [1] Hamadeh AO, Ingalls BP, Sontag ED, 2012. Dynamic phenotypes as criteria for model discrimination: fold-change detection in *R. sphaeroides*. arXiv:1204.1941v1 [q-bio.MN].
- [2] Monod J, Wyman J, Changeux JP, 1965. On the nature of allosteric transitions: A plausible model. *J Mol Biol* **12**:88118.
- [3] Shoval O, Goentoro L, Hart Y, Mayo A, Sontag E, and Alon U, 2010. Fold-change detection and scalar symmetry of sensory input fields. *Proc Natl Acad Sci USA* **107**:1599516000.
- [4] Lazova MD, Ahmed T, Bellomo D, Stocker R, Shimizu TS, 2011. Response rescaling in bacterial chemotaxis. *Proc Natl Acad Sci USA* **108**:1387013875.
- [5] Shoval O, Alon U, and Sontag ED, 2011. Symmetry invariance for adapting biological systems. *SIAM J Appl Dyn Syst*, **10**:857-886.
- [6] Kojadinovic M, Armitage JP, Tindall MJ, Wadhams GH, 2012. *Rhodobacter sphaeroides* chemotaxis response kinetics: Complexities in signalling but similarities in responses. Submitted.
- [7] Porter SL, Wadhams GH, Armitage JP, 2011. Signal processing in complex chemotaxis pathways. *Nat Rev Microbiol* **9**:153-65.
- [8] Barkai N, Leibler S, 1997. Robustness in simple biochemical networks. *Nature* **387**:913-917.
- [9] Yi TM, Huang Y, Simon MI, Doyle J, 2000. Robust perfect adaptation in bacterial chemotaxis through integral feedback control. *Proc Natl Acad Sci USA* **97**:46494653.
- [10] Sontag ED, 2003. Adaptation and regulation with signal detection implies internal model. *Syst Control Lett*, **50**:119-126.
- [11] Tu Y, Shimizu TS, Berg HC, 2008. Modeling the chemotactic response of *Escherichia coli* to time-varying stimuli. *Proc Natl Acad Sci USA* **105**:1485514860.
- [12] Roberts M, August E, Hamadeh A, Maini P, McSharry P, Armitage JP, Papachristodoulou A, 2009. A new approach for elucidating biological signalling networks. *BMC Syst Biol*, **3**:105.
- [13] Hamadeh A, Roberts M, August E, McSharry P, Maini P, Armitage JP, Papachristodoulou A, 2011. Feedback control architecture and the bacterial chemotaxis network. *PLoS Comp Biol* **7**:e1001130.
- [14] Porter SL, Armitage JP, 2002. Phosphotransfer in *Rhodobacter sphaeroides* chemotaxis. *J Mol Biol* **324**:3545.
- [15] Sourjik V, Berg HC, 2004. Functional interactions between receptors in bacterial chemotaxis. *Nature* **428**:437441.
- [16] Keymer JE, Endres RG, Skoge M, Meir Y, Wingreen NS, 2006. Chemosensing in *Escherichia coli*: Two regimes of two-state receptors. *Proc Natl Acad Sci USA* **103**:17861791.
- [17] Clausnitzer D, Oleksiuk O, Lvdok L, Sourjik V, Endres RG, 2010. Chemotactic Response and Adaptation Dynamics in *Escherichia coli*. *PLoS Comp Biol* **6**(5):e1000784.
- [18] Mello BA, Tu Y, 2005. An allosteric model for heterogeneous receptor complexes: Understanding bacterial chemotaxis responses to multiple stimuli. *Proc Natl Acad Sci USA* **102**:1735417359.
- [19] Shimizu TS, Tu Y, Berg HC, 2010. A modular gradient-sensing network for chemotaxis in *Escherichia coli* revealed by responses to time-varying stimuli. *Mol. Syst. Biol.* **6**:382